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Incubating Minds, Catalyzing Careers

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

20CS8504 & PROJECT WORK

END SEMESTER PRACTICAL EXAMINATION- APRIL / MAY - 2026

Batch No:5

Date: 30/3/2026

Session: FN

AI-ENABLED MORPHED IMAGE DETECTION AND DIGITAL FORENSIC ANALYSIS IN SECURE WEB SYSTEMS

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OBJECTIVE OF THE PROJECT

1. The primary objective of this project is to design and develop an intelligent system capable of detecting morphed and forged digital images using advanced image processing and digital forensic techniques.

2. The system aims to identify manipulated regions in images, such as copy-move and feature alteration forgeries, by extracting and analyzing key structural patterns effectively.

3. The project also focuses on preventing the spread of forged images by providing reliable forensic evidence, thereby enhancing security and trust in secure web and digital systems.

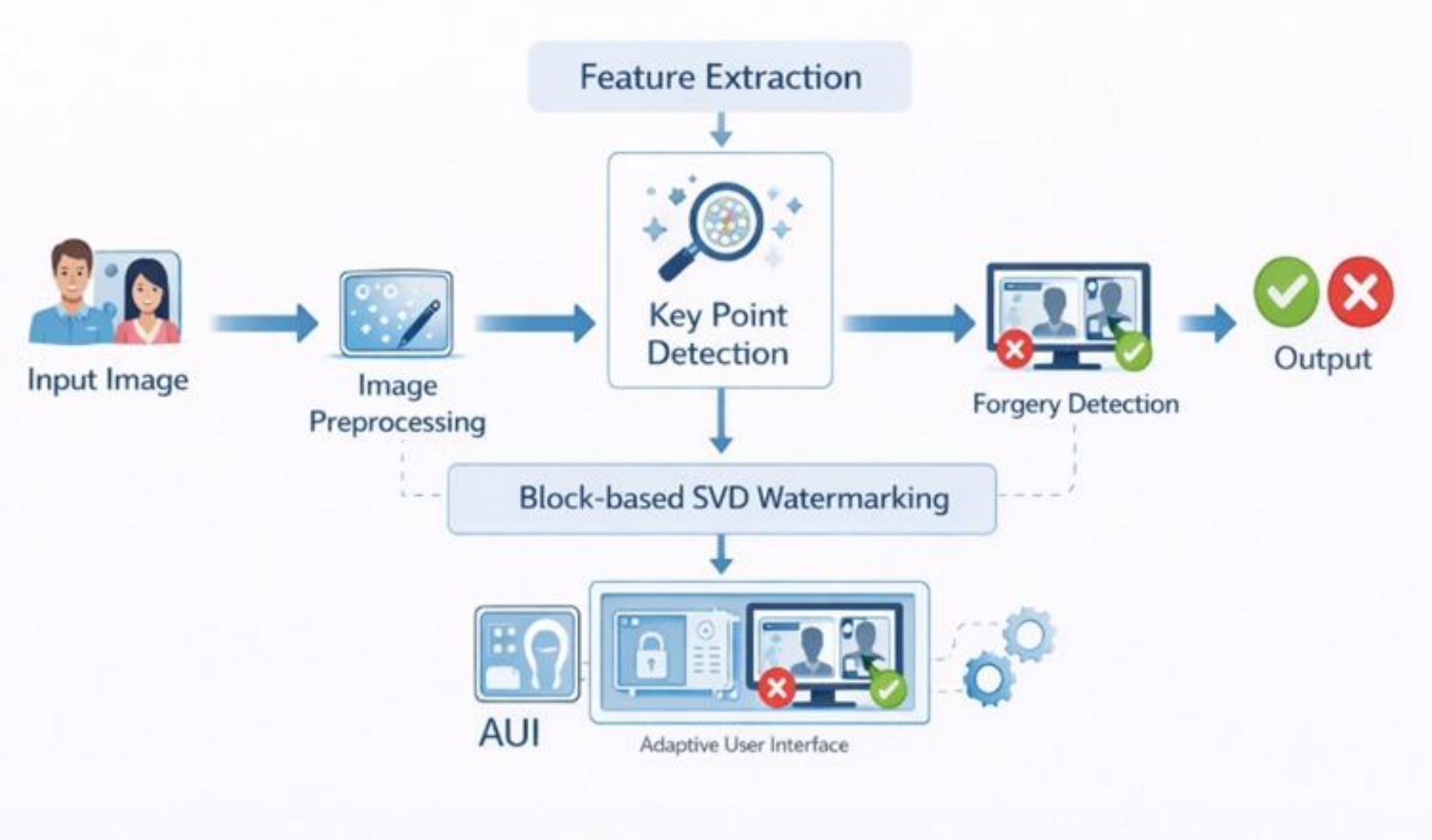
INTRODUCTION

1. Image and visual process have been updated where the digital image will be the information sources
2. The ambition behind the image content forgery is to perform the manipulations in a way, making them hard to reveal through the naked eye, and use these creations for malicious purposes
3. In recent years, an exciting field, digital image forensics, has emerged which finds the evidence of forgeries in digital images
4. Recent research found that attacks based on morphed face images, i.e. morphing attacks, pose a severe security risk to face recognition systems.
5. A reliable morphing attack detection from a single face image remains a research challenge since cameras and morphing techniques used by an attacker are unknown at the time of classification.

LITERATURE SURVEY

TITLE OF PAPER	AUTHOR(S)	PAPER GIST	TECHNOLOGY USED
A Survey of Digital Image Forgery Detection Techniques	Hany Farid	IEEE Signal Processing Magazine (2017)	Digital Image Forensics, Statistical Feature Analysis
Image Forgery Detection Using SURF Features	H. Bayram et al.	Elsevier – Signal Processing (2016)	SURF, Keypoint Detection, Feature Matching
Copy-Move Forgery Detection Using Block-Based Methods	J. Fridrich, D. Soukal	IEEE Transactions on Information Forensics (2018)	Block-Based Image Analysis, DCT, PCA
Deep Learning for Image Forgery ,Manipulation Detection	Y. Rao, J. Ni	IEEE Conference on Computer Vision(2020)	CNN, Deep Learning, Image Classification
Deep Learning for Image Forgery ,Manipulation Detection	Y. Rao, J. Ni	IEEE Conference on Computer Vision(2020)	CNN, Deep Learning, Image Classification
Secure Digital Image Forensic System for Web Applications	A. Singh, P. Kumar	Springer – International Journal(2019)	Secure Web Systems, Image Hashing, Forensic Logging
Improving SURF-Based Copy-Move Forgery Detection	M. Hussain et al.	IEEE International Conference(2020)	SISR, SURF, Image Preprocessing

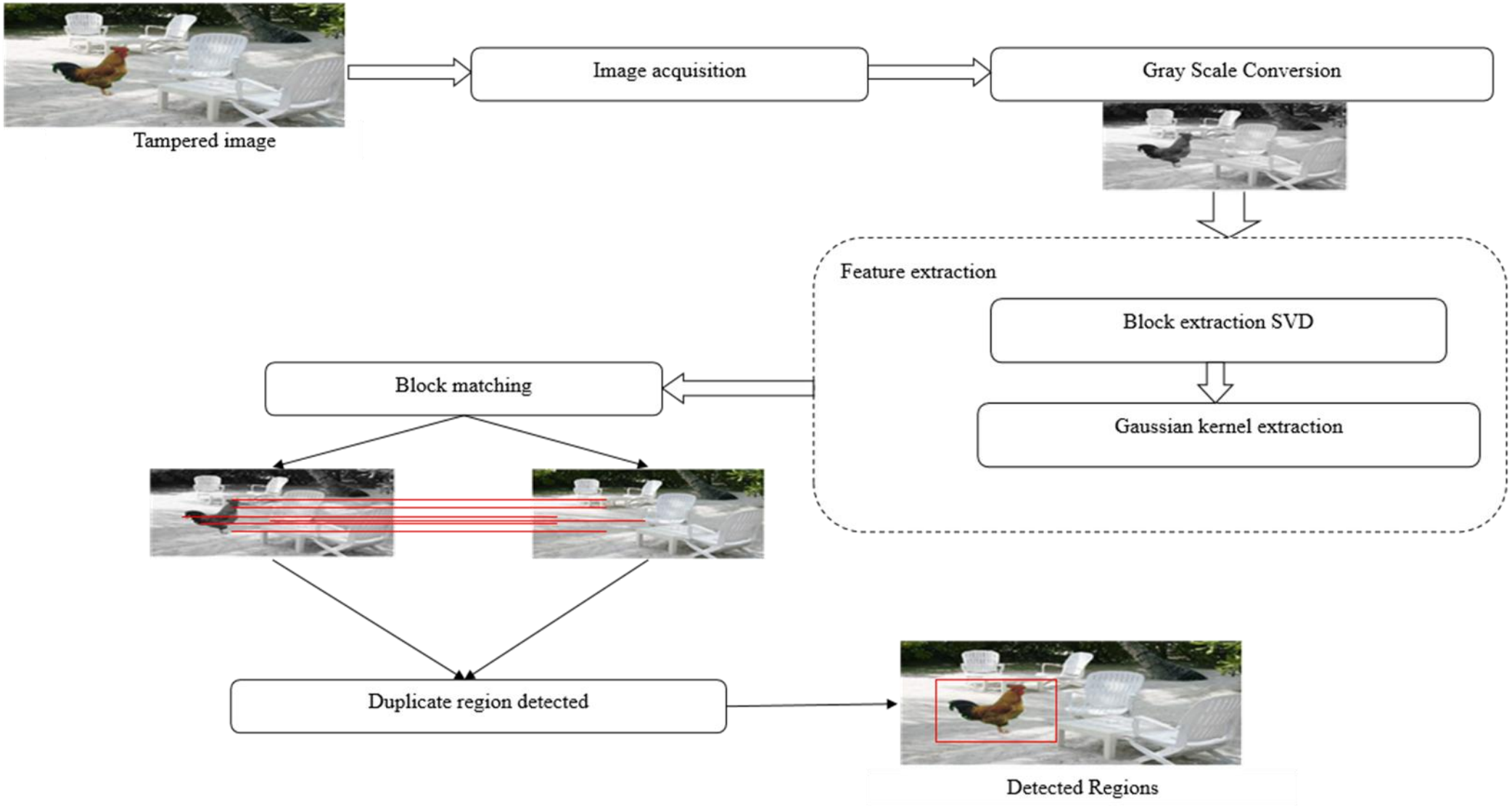
EXISTING SYSTEM ARCHITECTURE



PROBLEM IDENTIFIED

1. Rapid growth of digital image manipulation using advanced editing tools.
2. Morphed and forged images are difficult to detect with the human eye.
3. Face morphing attacks can bypass biometric and face recognition systems.
4. Traditional forensic techniques have limited accuracy in detecting complex image forgeries.
5. Need for an automated and intelligent system to detect manipulated images and ensure digital security.

PROPOSED SYSTEM ARCHITECTURE



SOFTWARE AND HARDWARE REQUIREMENTS

HARDWARE

1. **CORE PROCESSOR : 2.6.0GHZ**
2. **RAM : 1GB**
3. **HARD DISK : 160MB**

SOFTWARE

1. **FRONT END : JAVASCRIPT**
2. **BACK END : JAVA**
3. **DATABASE : MYSQL**

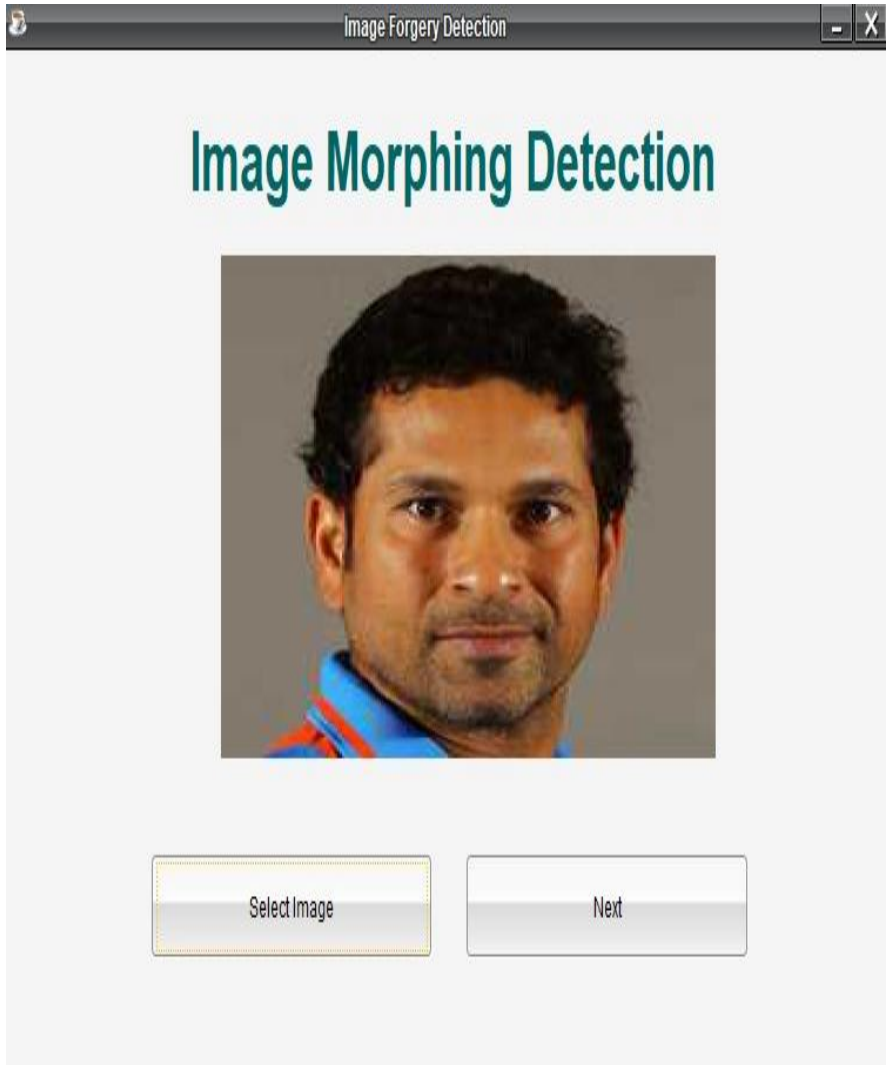
MODULES

1. Image Acquisition
2. Gray scale Conversion
3. Block extraction SVD
4. Gaussian Kernel PCA
5. Block Matching

MODULE-1
IMAGE ACQUISTION

1. The system captures or loads digital images from reliable sources and converts them into a machine-readable format

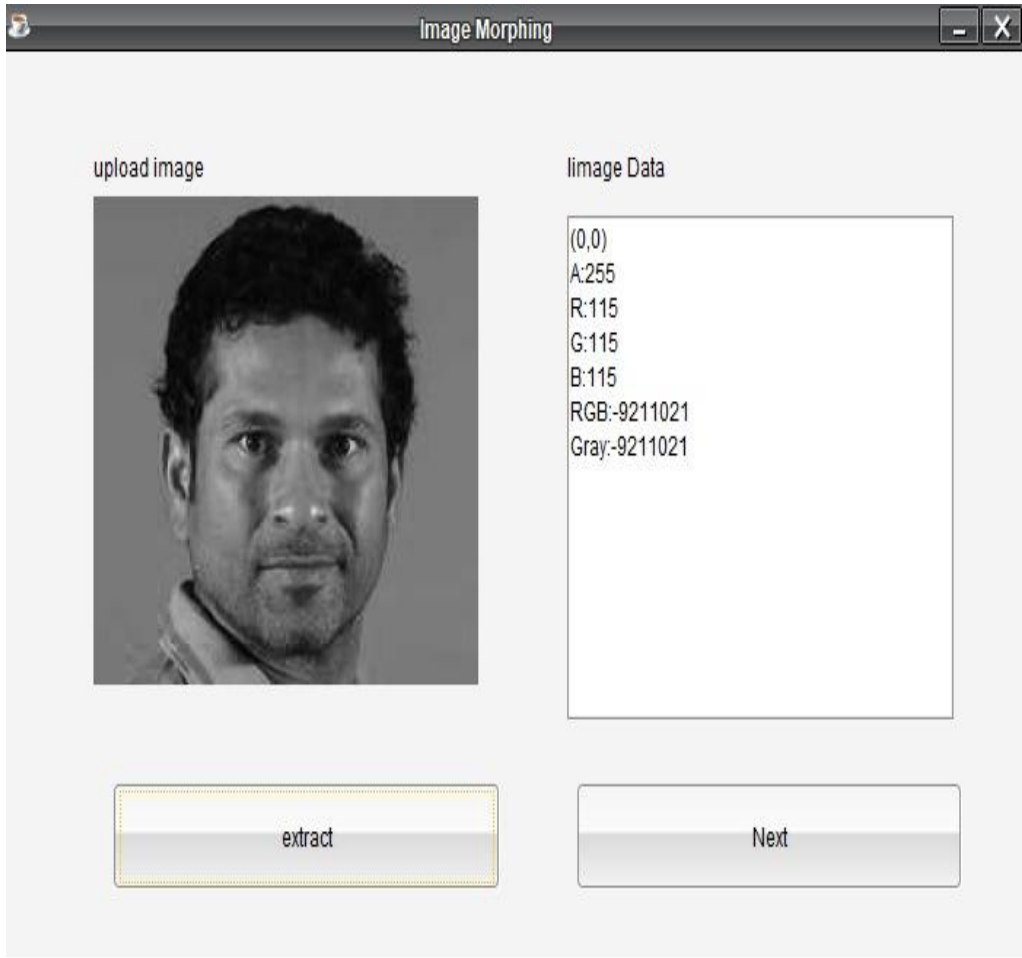
2. Basic preprocessing operations are applied to maintain image clarity and consistency. This step improves the accuracy of subsequent feature extraction and analysis.



MODULE-2
GRAY SCALE CONVERSION

1. The input color image is converted into a grayscale image to reduce dimensional complexity. This helps in faster processing while retaining essential visual information.

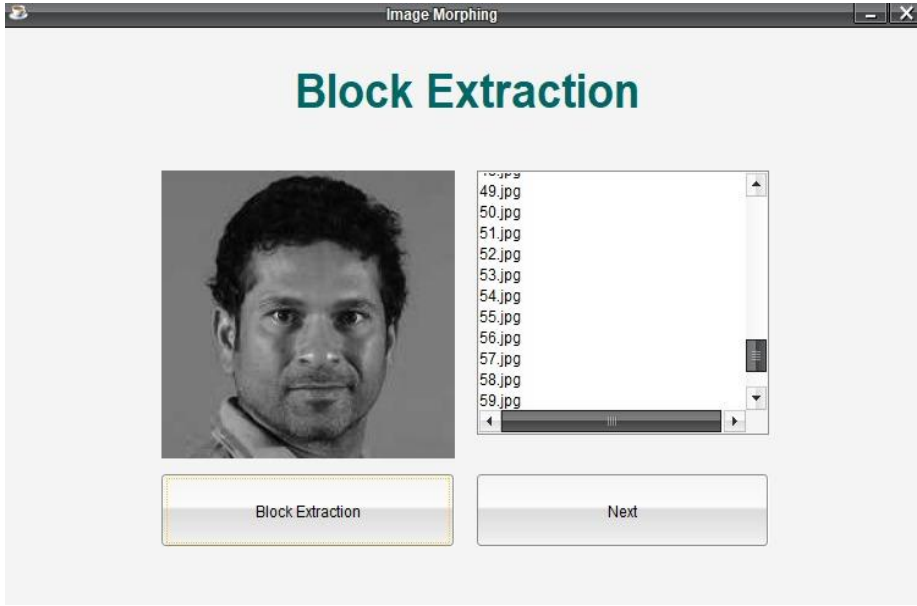
2. Grayscale conversion limits pixel values between 0 and 255. This normalization simplifies computations and enhances algorithm efficiency



MODULE-3
BLOCK EXTRACTION SVD

1. The grayscale image is divided into small overlapping blocks to capture local image characteristics. This allows effective analysis of fine-grained image regions.

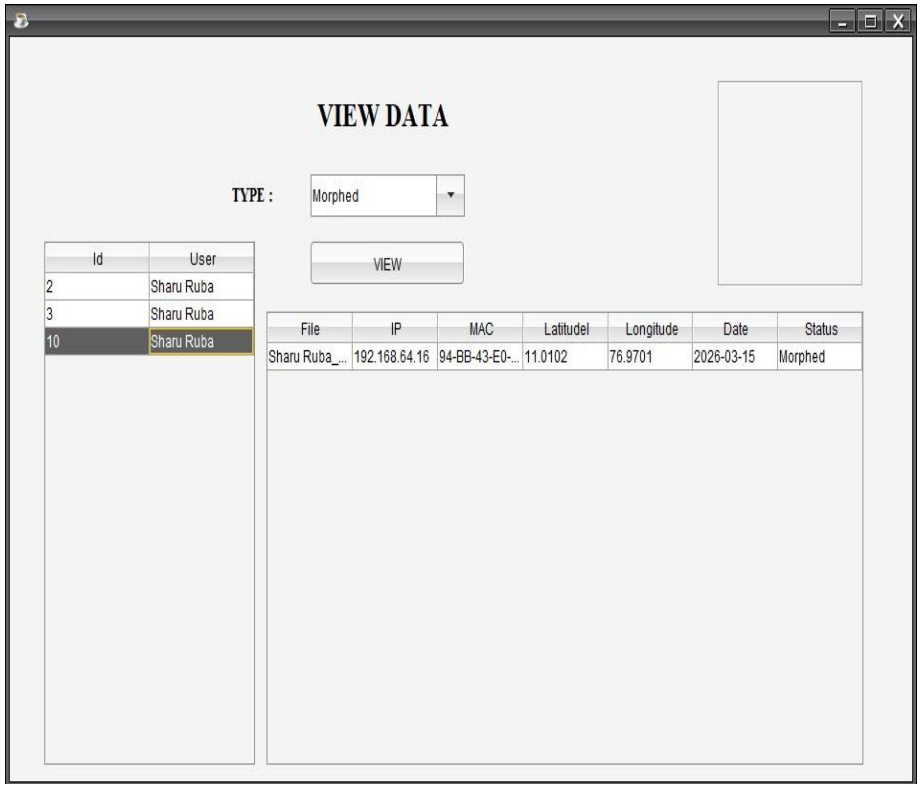
2. SVD is applied to each block to extract compact and robust feature vectors. These features remain stable even after image transformations



MODULE-4
GAUSSIAN KERNEL PCA

1. Gaussian Kernel PCA maps extracted features into a higher-dimensional space. This helps in capturing the non-linear relationships within image data given.

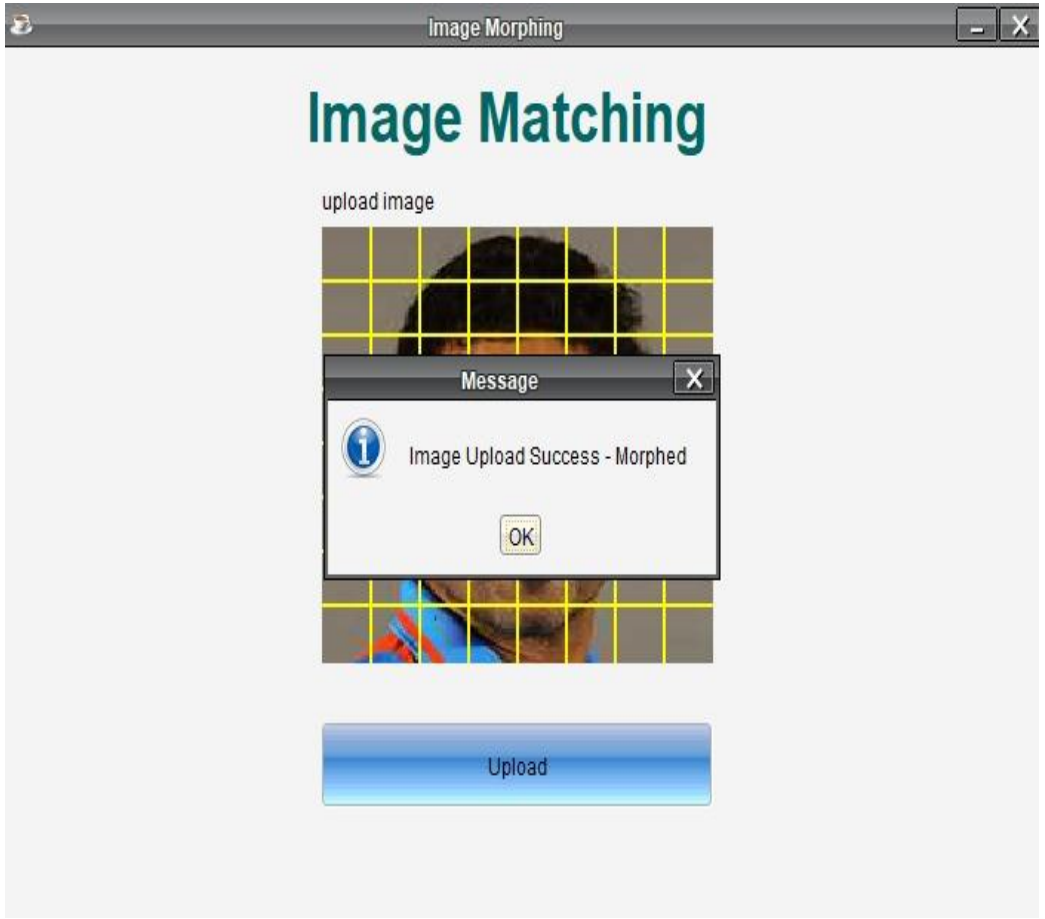
2. The transformed feature space improves separation between original and forged regions. This significantly increases detection accuracy for morphing attacks.



MODULE-5
BLOCK CHAIN MATCHING

1. Feature vectors from different image blocks are compared to identify similarities. Similar patterns indicate potential copy-move or morphed regions.

2. The system highlights duplicated or altered regions clearly. This provides strong forensic evidence even after resizing or cropping.



CONCLUSION

1. The proposed AI-enabled system effectively detects morphed and forged regions in digital images. It overcomes the limitations of existing manual detection methods by an improved detection accuracy of 93% in identifying morphed and forged images.

2. The system provides reliable forensic evidence to enhance security in digital platforms.It can be further improved using deep learning for real-time and large-scale applications.

FUTURE ENHANCEMENTS

1. AI-enabled morphed image detection and digital forensic analysis are rapidly evolving fields due to the increasing use of AI-generated images and deepfakes.

2. Modern synthetic media technologies based on neural networks such as Generative Adversarial Networks (GANs) create highly realistic fake images that are difficult to detect with traditional forensic tools.